

07:00 Fuza Mntungwa (student no: 2805001951)

07:30

08:00 Exercise 4: SQL Date Functions & Date Math

08:30

09:00 (01) ^{DAYNAME} SELECT ~~DAY~~ (order-date) AS day-name,
09:30 order-id,
10:00 customer-id,
10:30 order-date
11:00 FROM orders ;
11:30

12:00	order-id	customer-id	order-date	day-name
12:30	1001	101	2026-05-01	Friday
13:00	1002	102	2026-05-02	Saturday
13:30	1003	103	2026-05-03	Sunday
14:00	1004	104	2026-05-04	Monday
14:30	1005	105	2026-05-05	Tuesday

15:00

15:30 (02) SELECT customer-id,
16:00 customer-name,
16:30 signup-date,
17:00 ^{MONTHNAME} ~~MONTH~~ (signup-date) AS signup-month name
17:30 FROM customer-signups ;
18:00
18:30
19:00
19:30
20:00

"Do a little more each day than you think you possibly can."
Lowell Thomas

OCTOBER 2025	NOVEMBER 2025
S M T W T F S	S M T W T F S
1 2 3 4	1
5 6 7 8 9 10 11	2 3 4 5 6 7 8
12 13 14 15 16 17 18	9 10 11 12 13 14 15
19 20 21 22 23 24 25	16 17 18 19 20 21 22
26 27 28 29 30 31	23 24 25 26 27 28 29
	30

07:00	customer-id	customer-name	sign up date	sign up - month - name
07:30	201	John	2026-01-15	January
08:00	202	Mary	2026-02-20	February
08:30	203	Peter	2026-03-05	March
09:00	204	Sarah	2026-04-18	April
09:30	205	Thabo	2026-05-30	May

10:00

10:30

11:00 (03) SELECT sale_id,
 11:30 product-name,
 12:00 sale_date,
 12:30 MONTH (sale_date) AS sale-month
 13:00 FROM sales ;

13:30

14:00	sale_id	product-name	sale-date	sale-month
14:30	5001	Laptop	2026-01-10	1
15:00	5002	Mouse	2026-02-15	2
15:30	5003	Keyboard	2026-03-20	3
16:00	5004	Monitor	2026-04-25	4
16:30	5005	Desk	2026-05-30	5

17:00

17:30 (04) SELECT transaction-id,
 18:00 customer-id,
 18:30 transaction-date,
 19:00 ~~YEAR~~ YEAR (transaction-date) AS transaction-year;
 19:30 FROM transactions ;

20:00

DECEMBER 2025	JANUARY 2026
S M T W T F S	S M T W T F S
1 2 3 4 5 6	1 2 3
7 8 9 10 11 12 13	4 5 6 7 8 9 10
14 15 16 17 18 19 20	11 12 13 14 15 16 17
21 22 23 24 25 26 27	18 19 20 21 22 23 24
28 29 30 31	25 26 27 28 29 30 31

"If we could sell our experiences for what they cost us,
 we'd all be millionaires."
 Abigail Van Buren

	transaction-id	customer-id	transaction-date	transaction-year
07:00				
07:30	T001	101	2024-12-15	2024
08:00	T002	102	2025-01-20	2025
08:30	T003	103	2025-06-20	2025
09:00	T004	104	2026-02-05	2026
09:30	T005	105	2026-05-25	2026

(05) SELECT delivery-id,
customer-id,
delivery-date,
DAY(delivery-date) AS day-of-month
FROM deliveries;

	delivery-id	customer-id	delivery-date	day-of-month
14:00				
14:30	D001	101	2026-05-01	1
15:00	D002	102	2026-05-08	8
15:30	D003	103	2026-05-15	15
16:00	D004	104	2026-05-22	22
16:30	D005	105	2026-05-29	29

(06) SELECT employee-id,
employee-name,
department,
CURRENT_DATE() AS today-date
FROM employees;

"Work hard – Do your best, keep your word, never get too big for your britches, trust in God, have no fear and never forget a friend."

Harry S. Truman

OCTOBER 2025

S	M	T	W	T	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

NOVEMBER 2025

S	M	T	W	T	F	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30						

07:00	employee_id	employee_name	department	salary_date
07:30	301	Nandi	Sales	2026-06-24
08:00	302	Brian	IT	2026-06-24
08:30	303	Leat'o	Finance	2026-06-24
09:00	304	Sipho	HR	2026-06-24
09:30	305	Aisha	Marketing	2026-06-24

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12:30

13:00

13:30

(07) SELECT order_id,
customer_id,
order_date_text,
TO_DATE (order_date_text, 'dd-mm-yyyy')
FROM online_orders;

14:00	order_id	customer_id	order_date_text	order_date
14:30	4001	101	2026-01-15	15-JAN-2026
15:00	4002	102	2026-02-20	20-FEB-2026
15:30	4003	103	2026-03-25	25-MAR-2026
16:00	4004	104	2026-04-10	10-APR-2026
16:30	4005	105	2026-05-05	05-MAY-2026

17:00

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18:00

18:30

19:00

19:30

20:00

DECEMBER 2025	JANUARY 2026
S M T W T F S	S M T W T F S
1 2 3 4 5 6	1 2 3
7 8 9 10 11 12 13	4 5 6 7 8 9 10
14 15 16 17 18 19 20	11 12 13 14 15 16 17
21 22 23 24 25 26 27	18 19 20 21 22 23 24
28 29 30 31	25 26 27 28 29 30 31

"Life is about timing."
Carl Lewis

07:00

(08) SELECT payment-id,
customer-id,
payment-date,
TO_CHAR (payment-date, 'YYYY-MM-DD')
FROM payment-dates;

07:30

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20:00

payment-id	customer-id	payment-date	formatted-payment-date
p001	101	2026-01-05	'2026-01-05'
p002	102	2026-02-10	'2026-02-10'
p003	103	2026-03-15	'2026-03-15'
p004	104	2026-04-20	'2026-04-20'
p005	105	2026-05-25	'2026-05-25'

(09) SELECT customer-id,
customer-name,
last-purchase-date,
DATEDIFF (CURRENT_DATE(), last-purchase-date) AS
days-since-last-purchase
FROM customer-purchases;

customer-id	customer-name	last-purchase-date	days-since-last-purchase
501	John	2026-05-01	55
502	Mary	2026-05-10	46
503	Peter	2026-05-15	41
504	Sarah	2026-05-20	36
505	Thebo	2026-05-25	31

"First say to yourself what you would be; and then do what you have to do."
Epictetus

OCTOBER 2025	NOVEMBER 2025
S M T W T F S	S M T W T F S
1 2 3 4	1
5 6 7 8 9 10 11	2 3 4 5 6 7 8
12 13 14 15 16 17 18	9 10 11 12 13 14 15
19 20 21 22 23 24 25	16 17 18 19 20 21 22
26 27 28 29 30 31	23 24 25 26 27 28 29
	30

07:00 (10) SELECT order-id,
 07:30 customer-id,
 08:00 order-date,
 08:30 DATEADD (order-date, 7)
 09:00 FROM shipping-orders;
 09:30

	order-id	customer-id	order-date	expected-delivery-date
10:00				
10:30	6001	101	2026-05-01	2026-05-08
11:00	6002	102	2026-05-03	2026-05-10
11:30	6003	103	2026-05-05	2026-05-12
12:00	6004	104	2026-05-07	2026-05-14
12:30	6005	105	2026-05-09	2026-05-16

13:00
 13:30 (11) SELECT booking-id,
 14:00 customer-id,
 14:30 booking-date,
 15:00 YEAR (booking-date) AS booking-year,
 15:30 MONTH (booking-date) AS booking-month,
 16:00 DAY (booking-date) AS booking-day
 16:30 FROM bookings;
 17:00

	booking-id	customer-id	booking-date	booking-year	booking-month	booking-day
17:30						
18:00	B001	101	2026-01-12	2026	1	12
18:30	B002	102	2026-02-18	2026	2	18
19:00	B003	103	2026-03-22	2026	3	22
19:30	B004	104	2026-04-09	2026	4	09
20:00	B005	105	2026-05-27	2026	5	27

DECEMBER 2025	JANUARY 2026
S M T W T F S	S M T W T F S
1 2 3 4 5 6	1 2 3
7 8 9 10 11 12 13	4 5 6 7 8 9 10
14 15 16 17 18 19 20	11 12 13 14 15 16 17
21 22 23 24 25 26 27	18 19 20 21 22 23 24
28 29 30 31	25 26 27 28 29 30 31

"If you do work that you love, and the work fulfils you, the rest will come."
 Oprah Winfrey

07:00 (12) SELECT order-id,
 07:30 customer-id,
 08:00 order-date,
 08:30 YEAR (order-date) AS order-year,
 09:00 amount
 09:30 FROM yearly-orders
 10:00 WHERE order-year = 2026 ;
 10:30

	order-id	customer-id	order-date	order-year	amount
11:00					
11:30	7001	101	2024-12-15	2024	500
12:00	7002	102	2025-01-20	2025	1200
12:30	7003	103	2025-06-10	2025	800
13:00	7004	104	2026-02-05	2026	1500
13:30	7005	105	2026-05-25	2026	2000

14:00 (13) SELECT order-id,
 14:30 customer-id,
 15:00 order-date,
 15:30 MONTH (order-date) AS order-month,
 16:00 amount
 16:30 FROM monthly-orders
 17:00 WHERE order-month = 3 ;

	order-id	customer-id	order-date	order-month	amount
17:30					
18:00	8001	101	2026-01-15	1	500
18:30	8002	102	2026-02-20	2	1200
19:00	8003	103	2026-03-10	3	800
19:30	8004	104	2026-03-25	3	1500
20:00	8005	105	2026-05-30	5	2000

"The game of life is the game of boomerangs. Our thoughts, deeds and words
 return to us sooner or later, with astounding accuracy."
 Florence Shinn

OCTOBER 2025							NOVEMBER 2025						
S	M	T	W	T	F	S	S	M	T	W	T	F	S
				1	2	3	4						1
5	6	7	8	9	10	11		2	3	4	5	6	7
12	13	14	15	16	17	18		9	10	11	12	13	14
19	20	21	22	23	24	25		16	17	18	19	20	21
26	27	28	29	30	31			23	24	25	26	27	28
								30					29

07:00 (14) SELECT subscription_id,
 07:30 customer_id,
 08:00 start_date,
 08:30 LAST_DAY(start_date) AS month_end_date
 09:00 FROM subscriptions ;

subscription_id	customer_id	start_date	month_end_date
SUB001	101	2026-01-10	2026-01-31
SUB002	102	2026-02-15	2026-02-28
SUB003	103	2026-03-20	2026-03-31
SUB004	104	2026-04-25	2026-04-30
SUB005	105	2026-05-30	2026-05-31

13:00 (15) SELECT send_id,
 13:30 customer_id,
 14:00 send_date,
 14:30 DATE_TRUNC('month', send_date) AS month_start_date
 15:00 FROM campaign_sends ;

send_id	customer_id	send_date	month_start_date
C001	101	2026-01-12	2026-01-01
C002	102	2026-02-18	2026-02-01
C003	103	2026-03-22	2026-03-01
C004	104	2026-04-09	2026-04-01
C005	105	2026-05-27	2026-05-01

DECEMBER 2025	JANUARY 2026
S M T W T F S	S M T W T F S
1 2 3 4 5 6	1 2 3
7 8 9 10 11 12 13	4 5 6 7 8 9 10
14 15 16 17 18 19 20	11 12 13 14 15 16 17
21 22 23 24 25 26 27	18 19 20 21 22 23 24
28 29 30 31	25 26 27 28 29 30 31

"Live out of your imagination, not your history."
 Stephen Covey

07:00 (16) SELECT invoice_id,
 07:30 customer_id,
 08:00 invoice_date,
 08:30 TO CHAR (invoice_date, mm-yy) AS invoice_month-year
 09:00 FROM invoice_dates ;
 09:30

invoice_id	customer_id	invoice_date	invoice_month-year
INV001	101	2026-01-05	January 2026
INV002	102	2026-02-10	February 2026
INV003	103	2026-03-15	March 2026
INV004	104	2026-04-20	April 2026
INV005	105	2026-05-25	May 2026

13:00 (17) SELECT customer_id,
 13:30 customer_name,
 14:00 date_of_birth
 14:30 DATEDIFF (CURRENT_DATE(), date_of_birth) AS
 15:00 customer_age
 15:30 FROM customer_birthdays ;
 16:00

customer_id	customer_name	date_of_birth	customer_age
901	John	1998-05-10	28
902	May	1990-08-20	36
903	Peter	2002-03-15	24
904	Sarah	1985-12-01	41
905	Thabo	2000-07-30	26

19:30

20:00

"Do not let what you cannot do interfere with what you can do."
 John Wooden

OCTOBER 2025							NOVEMBER 2025						
S	M	T	W	T	F	S	S	M	T	W	T	F	S
				1	2	3	4						1
5	6	7	8	9	10	11		2	3	4	5	6	7
12	13	14	15	16	17	18		9	10	11	12	13	14
19	20	21	22	23	24	25		16	17	18	19	20	21
26	27	28	29	30	31			23	24	25	26	27	28
								30					

07:00 (18) SELECT order-id,
 07:30 customer-id,
 08:00 order-date,
 08:30 DAYNAME (order-date) AS day-name,
 09:00 CASE
 09:30 WHEN day-name = 'Saturday' OR day-name = 'Sunday'
 10:00 THEN 'Weekend'
 10:30 ELSE 'Weekday'
 11:00 END AS day-type
 11:30 FROM weekend-orders;

order-id	customer-id	order-date	day-name	day-type
9001	101	2026-05-01	Friday	Weekday
9002	102	2026-05-02	Saturday	Weekend
9003	103	2026-05-03	Sunday	Weekend
9004	104	2026-05-04	Monday	Weekday
9005	105	2026-05-05	Tuesday	Weekday

15:30 (19) SELECT transaction-id,
 16:00 customer-id,
 16:30 transaction-date,
 17:00 QUARTER (transaction-date) AS transaction-quarter,
 17:30 amount
 18:00 FROM quarterly-transactions;

DECEMBER 2025	JANUARY 2026
S M T W T F S	S M T W T F S
1 2 3 4 5 6	1 2 3
7 8 9 10 11 12 13	4 5 6 7 8 9 10
14 15 16 17 18 19 20	11 12 13 14 15 16 17
21 22 23 24 25 26 27	18 19 20 21 22 23 24
28 29 30 31	25 26 27 28 29 30 31

"You cannot plough a field by turning it over in your mind."
 Nolan Bushnell

07:00	transaction-id	customer-id	transaction-date	transaction-quarter	amount
07:30	Q001	101	2026-01-15	1	500
08:00	Q002	102	2026-03-20	1	1200
08:30	Q003	103	2026-04-10	2	800
09:00	Q004	104	2026-07-05	3	1500
09:30	Q005	105	2026-10-25	4	2000

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(20) SELECT order-id,
customer-id,
order-date,
DATEDIFF(CURRENT_DATE(), order-date) AS days-since-order,
amount
FROM recent-orders
WHERE days-since-order > 30 ;

15:00	order-id	customer-id	order-date	days-since-order	amount
15:30	R001	101	2026-04-01	85	500
16:00	R002	102	2026-04-15	71	1200
16:30	R003	103	2026-05-01	55	800
17:00	R004	104	2026-05-10	46	1500
17:30	R005	105	2026-05-25	31	2000

18:00

18:30

19:00

19:30

20:00

"Few people will have the greatness to bend history itself; but each of us can work to change a small portion of events, and in the total of all these acts will be written the history of this generation."

Robert F. Kennedy

OCTOBER 2025							NOVEMBER 2025						
S	M	T	W	T	F	S	S	M	T	W	T	F	S
				1	2	3	4						1
5	6	7	8	9	10	11		2	3	4	5	6	7
12	13	14	15	16	17	18		8	9	10	11	12	13
19	20	21	22	23	24	25		14	15	16	17	18	19
26	27	28	29	30	31			20	21	22	23	24	25
								26	27	28	29	30	

07:00 (B) SELECT customer_id,
 07:30 customer_name,
 08:00 last_purchase_date,
 08:30 DATEDIFF(CURRENT_DATE(), last_purchase_date) AS
 09:00 days_since_last_purchase,
 09:30 CASE
 10:00 WHEN days_since_last_purchase ≤ 30 THEN
 10:30 'Active Customer'
 11:00 WHEN days_since_last_purchase BETWEEN 31 AND
 11:30 90 THEN 'At Risk Customer'
 12:00 ELSE 'Inactive Customer'
 12:30 END AS customer_status
 13:00 FROM customer_recency;

	customer_id	customer_name	last_purchase_date	days_since_last_purchase	customer_status
14:00					
14:30	1001	John	2026-05-25	31	At risk customer
15:00	1002	Mary	2026-05-10	46	At risk customer
15:30	1003	Peber	2026-04-01	85	At risk customer
16:00	1004	Sardh	2026-02-15	102	Inactive customer
16:30	1005	Thebo	2025-12-20	187	Inactive customer
17:00					

17:30
 18:00
 18:30
 19:00
 19:30
 20:00

DECEMBER 2025	JANUARY 2026
S M T W T F S	S M T W T F S
1 2 3 4 5 6	1 2 3
7 8 9 10 11 12 13	4 5 6 7 8 9 10
14 15 16 17 18 19 20	11 12 13 14 15 16 17
21 22 23 24 25 26 27	18 19 20 21 22 23 24
28 29 30 31	25 26 27 28 29 30 31

"One thing you can't recycle is wasted time."
 Unknown